

CodeAlpha_Exploratory Data Analysis

Task 2

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
plt.figure()
sns.set()
```

<Figure size 640x480 with 0 Axes>

```
df = pd.read_excel("Online Retail Data Set after data cleaning.xlsx")
```

```
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom	
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom	
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	

```
questions = [
    "What are the top selling products?",
    "Which country generates the most revenue?",
    "What is the monthly sales trend?",
    "Who are the top customers?",
    "Are there any unusual quantities or prices in the dataset?"
]
```

```
for q in questions:
    print("-", q)
```

```
- What are the top selling products?
- Which country generates the most revenue?
- What is the monthly sales trend?
- Who are the top customers?
- Are there any unusual quantities or prices in the dataset?
```

```
df.shape
```

```
(536641, 8)
```

```
df.columns
```

```
Index(['InvoiceNo', 'StockCode', 'Description', 'Quantity', 'InvoiceDate',
       'UnitPrice', 'CustomerID', 'Country'],
      dtype='object')
```

```
df.dtypes
```

```
0
InvoiceNo      object
StockCode      object
Description     object
Quantity       int64
InvoiceDate    datetime64[ns]
UnitPrice      float64
CustomerID     float64
Country        object

dtype: object
```

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 536641 entries, 0 to 536640
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   InvoiceNo        536641 non-null object
1   StockCode       536641 non-null object
2   Description     535187 non-null object
3   Quantity        536641 non-null int64
4   InvoiceDate     536641 non-null datetime64[ns]
5   UnitPrice       536641 non-null float64
6   CustomerID     401604 non-null float64
7   Country         536641 non-null object
dtypes: datetime64[ns](1), float64(2), int64(1), object(4)
memory usage: 32.8+ MB
```

df.describe()

	Quantity	InvoiceDate	UnitPrice	CustomerID	
count	536641.000000	536641	536641.000000	401604.000000	
mean	9.620029	2011-07-04 08:57:06.087421952	4.632656	15281.160818	
min	-80995.000000	2010-12-01 08:26:00	-11062.060000	12346.000000	
25%	1.000000	2011-03-28 10:52:00	1.250000	13939.000000	
50%	3.000000	2011-07-19 14:04:00	2.080000	15145.000000	
75%	10.000000	2011-10-18 17:05:00	4.130000	16784.000000	
max	80995.000000	2011-12-09 12:50:00	38970.000000	18287.000000	
std	219.130156	NaN	97.233118	1714.006089	

df.isnull().sum()

```
0
InvoiceNo      0
StockCode      0
Description    1454
Quantity       0
InvoiceDate    0
UnitPrice      0
CustomerID    135037
Country        0

dtype: int64
```

```
df = df.dropna(subset=['CustomerID'])
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df = df.drop_duplicates()
```

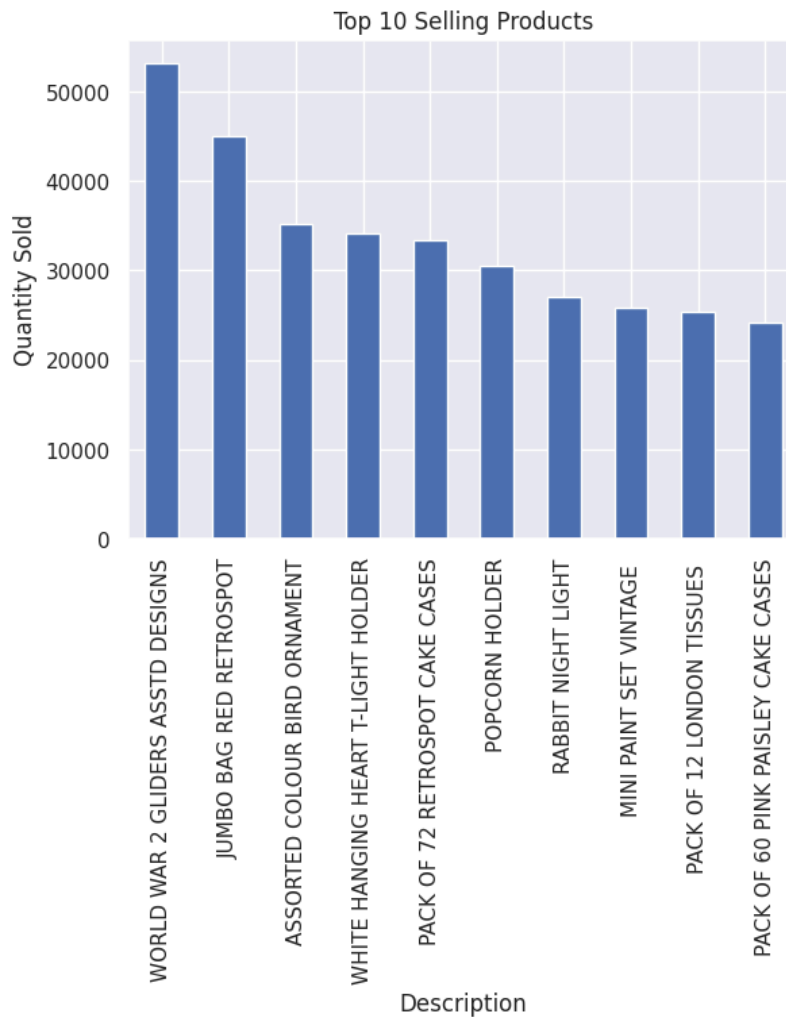
```
df["TotalPrice"] = df["Quantity"] * df["UnitPrice"]
```

```
df["InvoiceDate"] = pd.to_datetime(df["InvoiceDate"])
```

```
df["Year"] = df["InvoiceDate"].dt.year
df["Month"] = df["InvoiceDate"].dt.month
```

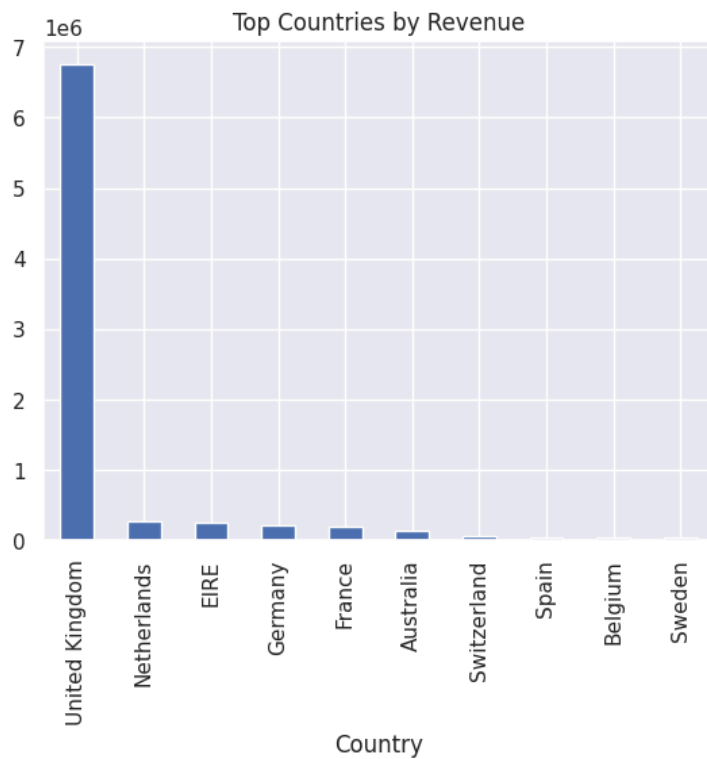
```
top_products = df.groupby("Description")["Quantity"].sum().sort_values(ascending=False).head(10)
```

```
plt.figure()
top_products.plot(kind="bar")
plt.title("Top 10 Selling Products")
plt.ylabel("Quantity Sold")
plt.show()
```



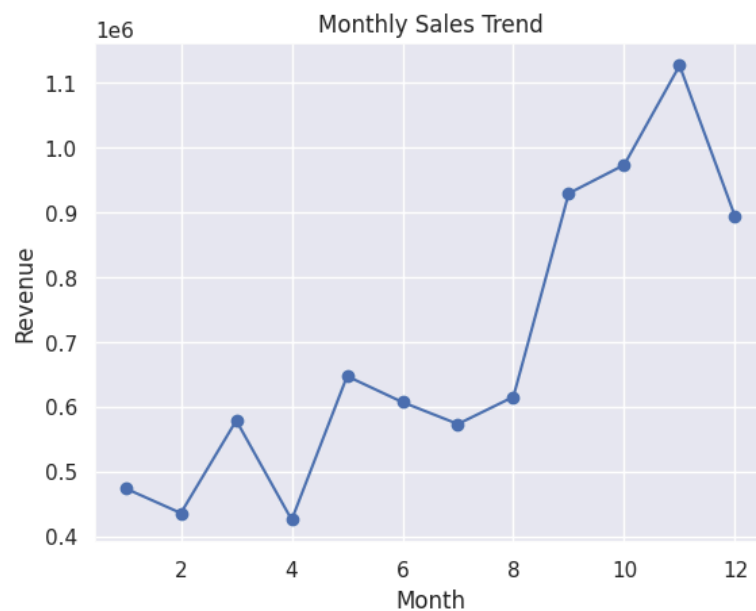
```
country_sales = df.groupby("Country")["TotalPrice"].sum().sort_values(ascending=False).head(10)
```

```
plt.figure()
country_sales.plot(kind="bar")
plt.title("Top Countries by Revenue")
plt.show()
```



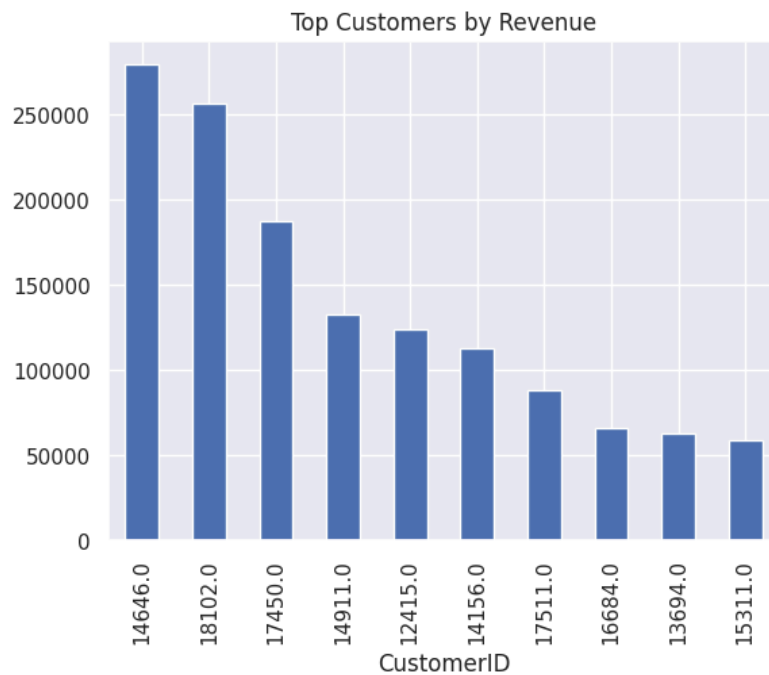
```
monthly_sales = df.groupby("Month")["TotalPrice"].sum()
```

```
plt.figure()
monthly_sales.plot(marker='o')
plt.title("Monthly Sales Trend")
plt.ylabel("Revenue")
plt.show()
```



```
top_customers = df.groupby("CustomerID")["TotalPrice"].sum().sort_values(ascending=False).head(10)
```

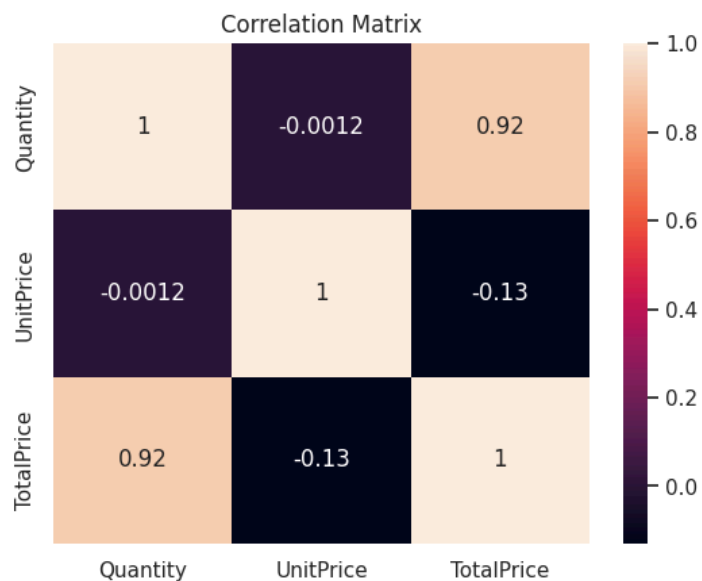
```
plt.figure()
top_customers.plot(kind="bar")
plt.title("Top Customers by Revenue")
plt.show()
```



```
plt.figure()
sns.scatterplot(x=df["Quantity"], y=df["TotalPrice"])
plt.title("Quantity vs Total Sales")
plt.show()
```



```
plt.figure()
sns.heatmap(df[["Quantity", "UnitPrice", "TotalPrice"]].corr(), annot=True)
plt.title("Correlation Matrix")
plt.show()
```



```
plt.figure()
sns.boxplot(x=df["Quantity"])
plt.title("Outliers in Quantity")
plt.show()
```



```
plt.figure()
sns.boxplot(x=df["UnitPrice"])
plt.title("Outliers in Unit Price")
plt.show()
```

